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Autoantibodies Testing for IgG and IgA in Selected PPMI samples utilizing the Sengenics Platform

This document was modified by Technome using documentation provided by Sengenics.

PPMI Project ID: [256]

Summary

Sengenics IgG and IgA autoantibody profiling was conducted on blood plasma of both Parkinson's Disease patients and healthy participants from the PPMI cohort. 127 PPMI Plasma samples from 48 subjects were run through the assays.

Acknowledgements

Data was generated utilizing funds generously contributed by AbbVie. Data was generated using the Sengenics Autoantibodies platform. AbbVie graciously provided data to both AMP-PD and LONI.





Associated Files

Normalized Protein Expression Files

“LONI_PPMI_Project_256_{Sample Tissue Type}_{Antibody Type}_{File Type}_2024-10-25.csv”

Sample Tissue Types: “Plasma”

File Types: “NetI” or “NCF”

Antibody Types: “IgG” or “IgA”

Column headers glossary

PROJECTID	Assigned by PPMI Study Team - three digit number
SAMPLEID	PP-number/10 digit Tube ID assigned by Biorepository
SMS_SUBJECT_ID	Internal Identifier ST Number (ST-00010000)
TESTNAME	Name of the test/analysis
TESTVALUE	Value generated by sample analysis
UNITS	Units of measure - may be blank
RUNDATE	Date the sample analysis was performed

List of Documents:

LONI_PPMI_Project_256_PLA_IgG_NetI_2024-10-25.csv

LONI_PPMI_Project_256_PLA_IgG_NCF_2024-10-25.csv

LONI_PPMI_Project_256_PLA_IgA_NetI_2024-10-25.csv

LONI_PPMI_Project_256_PLA_IgA_NCF_2024-10-25.csv

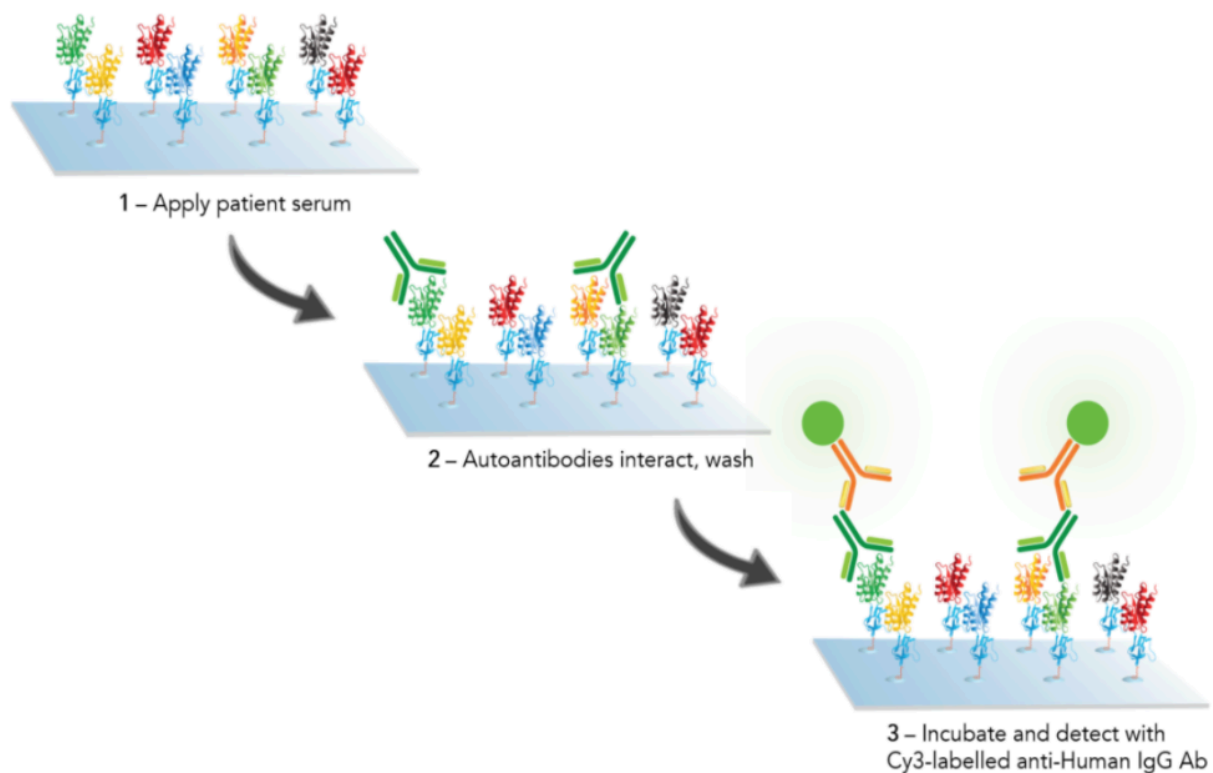




Method

Autoantibody assay

Samples were diluted with sample assay buffer (PBS, 0.1 % Triton, 0.1 % milk powder) at 1:200 and incubated for 1 hr at room temperature with agitation at 50 rpm. The bound antibody was detected using 1 µg/ml of anti-hIgG-Cy3. Autoantibody interactions against the antigens on each array were measured using an open format microarray laser scanner (Agilent Scanner) at a 10 µm resolution. Fluorescence levels were detected according to the manufacturer's instructions, whereby each spot is plotted using the Agilent Feature Extraction software. Each image was then saved as a 16-bit TIFF file.



Bioinformatics Methodology

a) Analysis of IgG Controls

To assess the binding capacity of the fluorescent-conjugated secondary antibody, we used the serial dilution of IgG on each array as a positive control. The benchmark that was used to





The following data pre-processing steps were performed before proceeding with data analysis:

Step 1:

Net intensities (NetI) for each spot were calculated by subtracting the background signal intensities from the foreground signal intensities of each spot. For each spot, the background signal intensity was calculated using a circular region with three times the diameter of the spot, centered on the spot (i.e. the local background).

Step 2:

For each antigen and control, the percentage of coefficient of variant (CV%) was calculated to determine the variations between the replica spots on each array.

$$CV\% = \frac{S.D.}{Mean} \times 100\%$$

Equation 2

If an antigen or control has a CV% > 20% and replica > 2, the replica spot which contributes to this high CV% value (i.e. outlier) was filtered out. For each antigen or control, this was done by calculating the absolute distance between the median value of the net intensities and individual net intensities of its replica spots. The spot with the highest absolute distance was then removed. CV% values were recalculated and the process repeated. Any antigens/controls with only 2 replicas remaining and CV% > 20% were flagged as "High CV". The mean NetI of these antigens/controls may be excluded from any downstream analysis.

Step 3:

For each antigen and control, the mean of the NetI of the remaining replica spots was calculated (i.e. the mean NetI).

d) Quality Control Analyses

IgG Controls

To assess the binding capacity of the fluorescent-conjugated secondary antibody, we used the serial dilution of IgG on each array as a positive control. The benchmark that was used to ensure that labeling efficiency and spot detection pass this QC parameter is based on a 3 series of 2-fold dilution of the IgG. The mean R-Squared (R²) between each IgG control and its experimental dilution is calculated across all IgG controls in all samples to assess the consistencies and linearity of the 3 IgG serial dilutions.





Cy3-BSA Controls

The Cy3-BSA controls on each array (Bovine serum albumin (BSA) protein pre-labelled with Cy3 prior to printing) act as housekeeping probes, with constant concentration of each Cy3BSA probe (1-3) spotted across all arrays and all slides. The mean CV% of Cy3-BSA replica spots is calculated by averaging the CV% of replica spots of each Cy3-BSA control across all 3 Cy3BSA controls in all samples. The Cy3-BSA controls passed the QC if the mean CV% value is $\leq 15\%$.

Overall Inter-replica and Intra-array CV%

Each antigen on the array is spotted in triplicate. To assess the accuracy of individual RFU we are reporting for this study, prior to performing any downstream analysis, the CV% between the replica spots of each antigen across all samples were calculated.

e) Downstream Analysis

i. Loess Normalisation

Before further analysis is performed on any given dataset, normalization of data is conducted to remove any systematic experimental variation in the dataset. In this study, a Loess normalization approach was applied across the dataset. The Loess normalization (Liu et al., 2019) method fits a nonlinear loess signal where equal weight is assigned to all data. The intensity-dependent bias is typically addressed within the representation of the data in the heatmap of net intensity. The method analyzes the whole dataset to estimate any bias or variation and attempts to eliminate any potential systemic variation or batch effect within the dataset. This method uses the `normalizeCyclicLoess` function which normalizes the columns of a matrix, cyclically applying Loess normalization to normalize each pair of columns to each other (Ballman et al., 2004).

ii. ComBat Batch Correction

ComBat allows users to adjust for batch effects in datasets where the batch covariate is known, using methodology described in (Johnson et al., 2007). It uses either parametric or non-parametric empirical Bayes frameworks for adjusting data for batch effects. Practical considerations limit the number of samples run at a given time, and replicate samples are generated in ways that introduce non-biological differences, or systematic "batch effects". For example, batch effects occur when adding replicates from different labs, array types, or platforms. In some cases, different lots of amplification reagent or the time of day of the assay have been demonstrated to cause batch effects. ComBat's Empirical Bayesian approach assumes phenomena resulting in batch effects affect many genes in similar ways and adjusts for these systematic batch biases common across genes. Users are returned an expression matrix that has been corrected for batch effects. The input data are assumed to be cleaned and normalized before batch effect removal





iii. Data Visualisation and Ordination

Ordination is a set of analyses that aim to summarize and present multivariate data to display differences between samples graphically in fewer dimensions than the original dataset. It is useful to relate samples to one another in relation to their underlying similarities (in this case, their autoantibody profiles). In this study, ordination techniques include performing unsupervised clustering (i.e. the clustering technique receives no input on any sample or clinical annotations e.g. "case" or "control") through heatmap and t- distributed stochastic neighbor embedding (tSNE) plots.

iv. Identification of samples with Polyspecific Autoantibody (PSA) Phenotype

The polyspecific autoantibody (PSA) phenotype, where certain individuals exhibit increased autoantibody signal across a substantial percentage of features on the i-Ome protein array was taken into consideration for clustering:

- i. This PSA phenomenon typically occurs in ~10% of individuals regardless of the disease status.
- ii. Possible causes of this may include recent infection and/or vaccination.
- iii. This phenotype was not enriched in a disease specific manner and therefore, the underlying signature would be excluded from downstream disease-of-interest subtyping to avoid PSA phenotype driving the clustering.
- iv. PSA status exists as a continuum, but for simplicity, samples are classified as PSA-positive or - negative, by unsupervised clustering with $k=2$.

v. Negative Control Filtration (NCF)

The NCF method is key in identifying true autoantibody signals, where antigens that lack true autoantibody signal are expected to be highly correlated with the negative control spots on the arrays (i.e. non-specific signal). This information can be used to select the antigens that display true autoantibody signals for downstream analyses. This particular method does not require knowledge of classes or groups of interest, e.g. "case" or "control", it simply excludes antigens that are highly correlated with the negative control spots. The NCF method utilizes Pearson's correlation, applied to the log₂ (net intensity) (pre-normalised) dataset. Correlation coefficients are calculated for each antigen against the negative control spot on the array (ZZ_con2). Antigens with the lowest correlation coefficients against the negative control spot are selected for downstream analyses. In this study, the default threshold selects a third of the data that is least correlated with the negative controls (this setting was chosen based on Sengenics' internal autoimmune datasets, where a relatively large proportion of the antigens exhibit true signal and





to avoid inadvertently excluding any true signal).

vi. Linear Modelling

The “Limma” package (Ritchie et al., 2015) was utilized to perform a differential expression analysis between Case Vs. Control across all 534 antigens. This analysis uses a moderated t-statistical method which calculates, for each antigen, the difference between two specified groups and determines the significance of these differences by measure of fold changes and p-values, respectively. The adjusted p- value uses Benjamini and Hochberg's method to adjust for multiple testing, to control the false discovery rate.

vii. Disease subtyping

Subtyping is performed on the subset of antigens deemed to have the highest biological relevance across the tested groups. The subtyping method was performed sequentially as follows: i) Using the NCF-selected antigens (Loess and ComBat normalized, row-scaled data), non-specific signal was approximated using the median across all antigens and all samples + 2 standard deviations (SD); ii) Antigens below this threshold were set to zero for filtering purposes; iii) Antigens that retained > 5% positivity across all samples were selected for subtyping; iv) Subtyping was performed on non-zeroed data (Loess normalised); v) Partitioning around Medoids (PAM) clustering was used with the most likely number of groups/clusters was based on the optimal silhouette width (Reynolds et al., 2006; Schubert & Rousseeuw, 2019). The optimal number of PAM clusters was selected as k=2.

viii. Biomarker ranking using a Receiver Operating Characteristic (ROC) curve analysis.

A ROC curve analysis was performed for each comparison group in this study across all 534 antigens. This method provides both visual and statistical means to identify biomarkers which are abundant in the case group but not in the controls. For each comparison and for each antigen, the true positive rate (TPR) or sensitivity and the false positive rate (FPR) or specificity at various threshold settings (Optimal Cut-off) were calculated using the normalized RFUs across all samples. For each antigen, the Area Under the Curve (AUC) was calculated, representing the ability of the model to classify/distinguish between the 2 analyzed groups. Stripcharts of the top 50 ranked antigens were also generated to visualize the distribution of autoantibody level for the comparison.

ix. Penetrance fold change

The penetrance fold change (pFC) method aims to detect true autoantibodies that differ between two groups of interest and was specifically designed for small sample size analyses.

The detailed steps involved are:

- i. Generate a per-antigen baseline based on a reference group (i.e.: for disease vs healthy, "healthy" is set as baseline/reference group) using appropriately normalized





input data.

- ii. For each antigen in each sample, calculate the individual fold changes based on the perantigen baseline value calculated in step (i).
- iii. For each group being compared, calculate per-antigen frequencies (penetrance frequency) of samples that exceed a fold change (FC) of 2 relative to the per-antigen baseline value calculated in step (i).
- iv. For samples that exceed the pFC defined in step (iii), calculate – on a per-antigen basis – the mean log2 intensity and the mean FC for each group of interest (included in the final results table).
- v. Using the frequencies calculated in step (iii), perform Fisher's exact testing on a per-antigen basis, comparing groups of interest ($p < 0.05$ significance threshold).
- vi. Determine antigens that are associated with PSA status by linear modeling ($p < 0.05$ & $FC > 1.1$) and report PSA-associated antigens as "yes"/"no" in the final results table.

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